AIX

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Approvals

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OS Upgrade the server

Introduction :

The server is running with the version of AIX 5.3 and If rootvg is mirrored, break the mirror and use one of the disks for the migration or assign a new SAN disk for this purpose

1. Unmirror the rootvg Duration: 90 Min

For example, server A is having the 2 disks i.e hdisk0 and hdisk1,so rootvg has mirrored(AIX5.3) break the mirroring and use the disk is for migration and we need to login into the server using HMC and follow these below steps.

1.Login the server using HMC console Duration : 5 Min

Login the server in putty session and check which HMC is connected to the server.

lsrsrc “IBM.MCP”

The above command displays which HMC is connected to our server. Next login server using the HMC console

2. Volumn Group Information Duration : 5 Min

Check the volumn group information .

lspv
lsvg rootvg
lspvg -l rootvg

3. Check the bootlist and check oslevel Duration : 5 Min

Check the bootlist and give the (AIX 5.3) hdisk0 first priority

bootlist -m normal -o
bootlist -m normal hdisk0 hdisk1
bootlist -m normal -o

Check the oslevel of the server.

#oslevel -s

4. Unmirror the rootvg Duration : 15 Min

For example server A is having the 2 disks i.e hdisk0 and hdisk1, rootvg has mirrored. Break the mirroring, to follow the below process

unmirrorvg rootvg hdisk1
reducevg -df rootvg hdisk1
lsvg -l rootvg

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Here we are reducing the hdisk1 from the rootvg

5. Clear the bootimage **Duration : 5 Min**

Clear the bootimage on unmirrored physical volume(hdisk1)
chpv -c hdisk1

6. Re-create the bootimage **Duration : 5 Min**

Check and re-create the bootimage on hdisk0. If server is reboot, it will bring up using the bootimage

bootlist -m normal -o
#bosboot -ad /dev/hdisk0
savebase

2. AIX Upgrade using NIMADM

The **nimadm** (Network Install Manager Alternate Disk Migration) is a utility that allows the system administrator to create a copy of **rootvg** to a free disk (or disks) and simultaneously migrate it to a new version or release level of AIX. The **nimadm** command uses NIM resources to perform this function.

There are several advantages to using the **nimadm** command over a conventional migration:

1. **Reduced downtime.** The migration is performed while the system is up and functioning normally. There is no requirement to boot from install media, and the majority of processing occurs on the NIM master.

2. The **nimadm** command **facilitates quick recovery in the event of migration failure**. Since the **nimadm** command uses **alt_disk_install** to create a copy of **rootvg**, all changes are performed to the copy (altinst_rootvg). In the even of serious migration installation failure, the failed migration is cleaned up and there is no need for the administrator to take further action. In the event of a problem with the new (migrated) level of AIX, the system can be quickly returned to the pre-migration operating system by booting from the original disk.

3. **The nimadm command allows a high degree of flexibility and customization in the migration process. This is done with the use of optional NIM customization resources: image_data, bosinst_data, exclude_files, pre-migration script, installp_bundle, and post-migration script.**

The nimadm command performs a migration in 12 phases:

1.Master issues alt_disk_install to the client. Makes a copy of the clients rootvg to target disk. In

this phase, the alternate root volume group (altinst_rootvg) is created.

2.Master creates the cache file systems in the nimadmvg volume group. Some initial checks for the required migration disk space are performed**.**

3 The NIM master copies the NIM client's data to the cache file systems i nimadmvg. This data copy is done via rsh.

4.If a pre-migration script resource has been specified, it i executed in this phase.

5.System configuration files are saved. Initial migration space is calculated and appropriate file system expansions are made The bos image is restored and the made.device database is merged (similar to a conventional migration).

6.All system filesets are migrated using installp Any required RPM images are also installed during this phase.

7.If a post-migration script resource has been specified, it is executed at this time.

8.The bosboot command is run to create a client boot image, which is written to the client's alternate boot logical volume (alt_hd5).

9.All the migrated data is now copied from the NIM master's local cache file and synced to the client's alternate rootvg via rsh.

10.The NIM master cleans up and removes the local cache file systems.

11.The alt_disk_install command is called again to make the final adjustments and put altinst_rootvg to sleep. The bootlist is set to the target disk.

12.Cleanup is executed to end the migration.

2.1 Pre-checks for the Client server Duration : 90 Min

Before migrating the server we can perform the pre-check in the client server

1. Check the oslevel
Duration : 1 Min

Verify the oslevel of the client server
oslevel -s

```
root@ukcmuxxa329:/ # oslevel -s
5300-12-05-1140
root@ukcmuxxa329:/ #
```

2. Client and NIM server communicated or not Duration : 5 Min

Check client server and NIM server are communicated or Not .If not add the entry in the /etc/hosts file on both the servers(client and NIM server)

ping <nim server ip>

```
root@ukcmuxxa329:/ # ping 10.4.211.57
PING 10.4.211.57: (10.4.211.57): 56 data bytes
64 bytes from 10.4.211.57: icmp_seq=0 ttl=254 time=0 ms
64 bytes from 10.4.211.57: icmp_seq=1 ttl=254 time=0 ms
64 bytes from 10.4.211.57: icmp_seq=2 ttl=254 time=0 ms
```

It is not commnicated add the entry in the **/etc/hosts** file
vi /etc/hosts
<target server hostname> <IP Address>
ping 10.4.211.57

3. Check the rootvg mirrored or not
Duration : 5 Min

Check the rootvg mirrored or not. Ensure the server has a spare disk (not allocated to a volume group) large enough space to contain a complete copy of its rootvg.

#lspv

```
root@ukcmuxxa329:/ # lspv
hdisk0          00c9d10e3fd7e5a6                    rootvg          active
hdisk1          00c9d10e27feb91d                    altinst_rootvg  active
hdisk2          0002c7d1de871d00                    discovg         active
hdisk3          00c9d10e4a09bc61                    cln_findbavg    active
hdisk4          0002c7d13f87d993                    cln_fincdcvg    active
hdisk5          00c9d10e4a09bcd2                    cln_findbavg    active
hdisk6          00c9d10e4a09bd42                    cln_findbavg    active
hdisk7          00c9d10ed9ce827f                    fincdcvg        active
hdisk8          00c9d10ed9ce82fd                    fincdcvg        active
hdisk9          00c9d10ed9ce88ad                    fincdcvg        active
hdisk10         00c9d10ec210035b                    None
hdisk11         00c9d10ec21003fa                    None
hdisk12         00c9d10ec2100496                    None
```

lsvg -l rootvg

```
root@ukcmuxxa329:/ # lsvg -l rootvg
rootvg:
LV NAME             TYPE       LPs     PPs     PVs  LV STATE      MOUNT POINT
hd5                 boot       1       1       1    closed/syncd  N/A
hd11admin           jfs2       1       1       1    open/syncd    /admin
paging00            paging     64      64      1    open/syncd    N/A
hd8                 jfs2log    1       1       1    open/syncd    N/A
hd4                 jfs2       23      23      1    open/syncd    /
hd2                 jfs2       36      36      1    open/syncd    /usr
hd9var              jfs2       16      16      1    open/syncd    /var
hd3                 jfs2       1       1       1    open/syncd    /tmp
hd1                 jfs2       5       5       1    open/syncd    /home
hd10opt             jfs2       28      28      1    open/syncd    /opt
lg_dumplv           sysdump    16      16      1    open/syncd    N/A
```

4. Break mirror copy of the rootvg or altinst_rootvg image **Duration: 15 Min**

If the client server having mirror copy of the rootvg first we need to break it. or if it is having altinst_rootvg, remove it

a. If rootvg mirrored ,break the mirroring

```
# unmirrorvg rootvg hdisk1
# reducevg rootvg hdisk1
# chpv -c hdisk1
```

(OR)

b. altinst_rootvg is in hdisk1 remove it and free the disk.

```
# alt_disk_install  -X hdisk1
# chpv -c hdisk1
```

5.To verify minimum Firmware level required for migration activity Duration : 5 Min

Verify the supported firmware levels

```
# lsmcode -c
```

```
[ukcmuxxa329]:/ >lsmcode -c
The current permanent system firmware image is EM350_038
The current temporary system firmware image is EM350_038
The system is currently booted from the temporary firmware image.
```

6. unlimit the file size for root user
Duration : 5 Min

We have to unlimit the file size for root user. Take the backup of /etc/security/limits

Just add the entry under the root user

fuser = -1

Then check **#ulimit -a** (command once)

```
# ulimit -a
```

```
[ukcmuxxa329]:/ >ulimit -a
time(seconds)        unlimited
file(blocks)         unlimited
data(kbytes)         unlimited
stack(kbytes)        4194304
memory(kbytes)       32768
coredump(blocks)     unlimited
nofiles(descriptors) 65536
threads(per process) unlimited
processes(per user)  unlimited
```

7. Check the missing filesets
Duration : 5 Min

Verify the missing filesets or un-installed filesets

```
# lppchk -vm3
# installp -C
```

8. Check the kernel size ,if it is in 32-bit change it into 64-bit kernal Duration : 30 Min

Check the kernel size,if it is in 32-bit we have to change it into the 64-bit . AIX 6.1 supports 64-bit kernel .if server is on the 64-bit then we no need to change the kernel. By using the following command we can find the kernel size .

prtconf -k

If this server is in 32-bit we have to follow the following process to change it into the 64-bit .
#ln -sf /usr/lib/boot/unix_64 /unix
#ln -sf /usr/lib/boot/unix_64 /usr/lib/boot/unix
#lslv -m hd5
#bosboot -ad /dev/ipldevice
#shutdown -Fr

9. Verify take performance and kernel parameters before the migration Duration : 10 Min

Verify the performance parameter and take the outputs, after migrating the server we have to validate both the parameters

lsattr -El sys0 – Check the maxproc value
vmo -a

Here check the parameters like maxperm,minperm,kernel_heap_psize,kernel_psize,maxfree ...etc

10. Take the mksysb backup of the server Duration : 30 Min

Take the backup of operating system of AIX 5.3 without any failures using mksysb or **alt_disk_install**

a. Before taking the backup if you need to exclude the files ,which you don't need to backup and add the entry in /etc/exclude.rootvg

^./tmp/prasadn/*

any directories or files contained under the /tmp/prasadn , backup will not taken

Then take rootvg backup using mksysb command and copy to the NIM server
#mksysb -ie /<directory or filesystem>/<Name of the file>

```
root@ukcmuxxa329:/home/prasadn #  mksysb -ie /home/prasadn/329mksysb/329backup

Creating information file (/image.data) for rootvg.

Creating list of files to back up.
Backing up 109812 files.............................
16025 of 109812 files (14%).............................
35079 of 109812 files (31%).............................
48932 of 109812 files (44%).............................
76149 of 109812 files (69%).............................
98826 of 109812 files (89%).............

109812 of 109812 files (100%)
0512-038 mksysb: Backup Completed Successfully.
```

(OR)

b. Alternate disk backup using alt_disk_install command

alt_disk_install -BC <hdiskN>

11. Verify the disk sizes **Duration : 2 Min**

Check the physical volumn sizes of hdisk0 and hdisk1 . both disks have enough free space for the NIMADM

bootinfo -s hdisk0
bootinfo -s hdisk1

```
root@ukcmuxxa329:/ # bootinfo -s hdisk0
30720
root@ukcmuxxa329:/ # bootinfo -s hdisk1
30720
root@ukcmuxxa329:/ #
```

12. Verify which NIM server is connected to our server **Duration : 5 Min**

cat /etc/niminfo

```
root@ukcmuxxa329:/ # cat /etc/niminfo
#------------------ Network Install Manager ---------------
# warning - this file contains NIM configuration information
#       and should only be updated by NIM
export NIM_NAME=ukcmuxxa329
export NIM_HOSTNAME=ukcmuxxa329.qbe.eo
export NIM_CONFIGURATION=standalone
export NIM_MASTER_HOSTNAME=uxxs1354
export NIM_MASTER_PORT=1058
export NIM_REGISTRATION_PORT=1059
export NIM_SHELL="shell"
export NIM_MASTERID=0002C78BD600
export NIM_BOS_IMAGE=/SPOT/usr/sys/inst.images/installp/ppc/bos
export NIM_BOS_FORMAT=rte
export NIM_HOSTS=" 127.0.0.1:loopback:localhost  10.4.210.54:ukcmuxxa329.qbe.eo
 10.4.211.57:uxxs1354 "
export NIM_MOUNTS=""
export ROUTES=" default:0:10.4.210.1 "
```

13. Copy /etc/mail/sendmail.cf to your home directory **Duration : 1 Min**

14. LV names are less than 11 characters
Duration : 2 Min

Make sure lv names are less than 11 characters if not nimadm will fail in phase1.

#lsvg -l rootvg

```
root@ukcmuxxa329:/ # lsvg -l rootvg
rootvg:
LV NAME             TYPE       LPs     PPs     PVs  LV STATE      MOUNT POINT
hd5                 boot       1       1       1    closed/syncd  N/A
hd11admin           jfs2       1       1       1    open/syncd    /admin
paging00            paging     64      64      1    open/syncd    N/A
hd8                 jfs2log    1       1       1    open/syncd    N/A
hd4                 jfs2       23      23      1    open/syncd    /
hd2                 jfs2       36      36      1    open/syncd    /usr
hd9var              jfs2       16      16      1    open/syncd    /var
hd3                 jfs2       1       1       1    open/syncd    /tmp
hd1                 jfs2       5       5       1    open/syncd    /home
hd10opt             jfs2       28      28      1    open/syncd    /opt
lg_dumplv           sysdump    16      16      1    open/syncd    N/A
```

15. Run the system information script **Duration : 20 Min**

We have to take the all the output of the server. Run the system information script and verify the outputs of the script

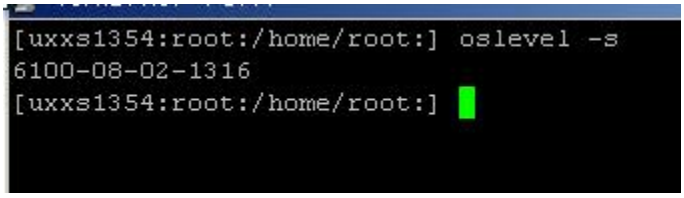
2.2 Pre-check for NIM Server **Duration :**

30 Min

The NIM server (as always) should be at the same or higher AIX level than the level you are migrating to on the client**.** Before migrating the server we can perform the pre-check in the NIM server.

15 . Check NIM server oslevel **Duration : 1 Min**

Check NIM server oslevel (uxxs1354)
#oslevel -s



The NIM server is running with oslevel of AIX 6.1 TL 08

16. Verify NIM resources
Duration: 5 Min

Verify the NIM resources on which level have in the server

lsnim

```
[uxxs1354:root:/home/root:] lsnim
master                          machines        master
boot                            resources       boot
nim_script                      resources       nim_script
net_10_4_211                    networks        ent
lpp_aix530                      resources       lpp_source
spot_aix530                     resources       spot
Global_Vio                      resources       fb_script
Global_Vio_Migration            resources       fb_script
Global_Customization            resources       fb_script
```

lsnim -c machines

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```
[uxxs1354:root:/home/root:] lsnim -c machines
master          machines        master
ukcmuxxd326     machines        standalone
uxxd1480        machines        standalone
uxss1311        machines        standalone
uxss1312        machines        standalone
uxxd1717        machines        standalone
uxxd1745        machines        standalone
uxxd1772        machines        standalone
uxxd1783        machines        standalone
uxxd1784        machines        standalone
uxpd1785        machines        standalone
uxpd1786        machines        standalone
uxxd1787        machines        standalone
uxxd1788        machines        standalone
ukcmuxxa329     machines        standalone
ukcmuxxa330     machines        standalone
[uxxs1354:root:/home/root:]
```

#lsnim -l resources

```
[uxxs1354:root:/home/root:] lsnim -c resources
boot                            resources       boot
nim_script                      resources       nim_script
lpp_aix530                      resources       lpp_source
spot_aix530                     resources       spot
Global_Vio                      resources       fb_script
Global_Vio_Migration            resources       fb_script
Global_Customization            resources       fb_script
Global_Custom                   resources       script
Global_Update                   resources       script
Global_Migration                resources       fb_script
Global_PreMigration_61          resources       script
Global_PreMigration_71          resources       script
53_bosmigr_32_jfs               resources       bosinst_data
53_bosmigr_64_jfs               resources       bosinst_data
```

17. Create the NIM resources if they are not in the NIM server

If resources are not being present in NIM server we have to define the lpp_source and spot for target oslevel to which level you want to migrate. Target client must registered with NIM server as a standalone NIM client

smitty nim

18. Check the lpp_source and spot ready or not **Duration : 2 Min**

Check the lpp_source (simages=yes) and SPOT Rstate= ready for use .

NIM server contains the lpp_source and spot AIX 6.1 TL 07, upto this oslevel only we are migrating the server.

lsnim -l lppsource_61-07-01

```
[uxxs1354:root:/home/root:] lsnim -l lppsource_61-07-01
lppsource_61-07-01:
   class       = resources
   type        = lpp_source
   arch        = power
   Rstate      = ready for use
   prev_state  = unavailable for use
   location    = /export/lpp_source/core/6.1/61-07-01
   simages     = yes
   alloc_count = 0
   server      = master
[uxxs1354:root:/home/root:]
```

lsnim -l spot_61-07-01

```
[uxxs1354:root:/home/root:] lsnim -l spot_61-07-01
spot_61-07-01:
   class          = resources
   type           = spot
   plat_defined   = chrp
   arch           = power
   Rstate         = ready for use
   prev_state     = verification is being performed
   location       = /export/spot/6.1/spot_61-07-01/usr
   version        = 6
   release        = 1
   mod            = 7
   oslevel_r      = 6100-07
   alloc_count    = 0
   server         = master
   if_supported   = chrp.64 ent
   Rstate_result  = success
[uxxs1354:root:/home/root:]
```

19 .List the disk sizes in NIM server **Duration : 5 Min**

Check the how many disks in the NIM server and ensure the NIM server has a spare disk to create a volume group (for example, migvg) with enough free space to cater for a complete copy of the client's rootvg.

lspv

```
[uxxs1354:root:/tmp:] lspv
hdisk0          0002c78bf7a69017                    rootvg          active
hdisk1          0002c78b71c44b66                    migvg           active
```

migvg is already created in NIM server, no need to create the new VG.

Check the hdisk sizes in the NIM server
bootinfo -s hdisk0

bootinfo -s hdisk1

```
[uxxs1354:root:/home/root:] bootinfo -s hdisk0
70006
[uxxs1354:root:/home/root:] bootinfo -s hdisk1
70006
```

20. Verify the bos.alt_disk_install fileset
Duration: 2 Min

To ensure that bos.alt_disk_install.rte fileset installed on NIM Server and also in the SPOT.If not we have to add the fileset to SPOT.

lslpp -l bos.alt_disk_install.rte

```
[uxxs1354:root:/home/root:] lslpp -l bos.alt_disk_install.rte
  Fileset                      Level  State      Description
  ----------------------------------------------------------------------------
Path: /usr/lib/objrepos
  bos.alt_disk_install.rte   6.1.8.15  COMMITTED  Alternate Disk Installation
                                                  Runtime
                                       EFIXLOCKED

Path: /etc/objrepos
  bos.alt_disk_install.rte   6.1.8.15  COMMITTED  Alternate Disk Installation
                                                  Runtime
                                       EFIXLOCKED
```

nim -o showres lppsource_61-07-01 | grep -i bos.alt_disk_install.rte

```
[uxxs1354:root:/home/root:] nim -o showres lppsource_61-07-01|grep -i bos.alt_disk_install.rte
  bos.alt_disk_install.rte    6.1.0.0                   I  N usr
  bos.alt_disk_install.rte    6.1.7.0                   S  N usr,root
  bos.alt_disk_install.rte    6.1.7.1                   S  N usr,root
  bos.alt_disk_install.rte    6.1.8.15                  S  N usr,root
[uxxs1354:root:/home/root:]
```

nim -o showres spot_61-07-01 | grep -i bos.alt_disk_install.rte

```
[uxxs1354:root:/home/root:] nim -o showres spot_61-07-01|grep -i bos.alt_disk_install.rte
  bos.alt_disk_install.rte  6.1.8.15    A     F    Alternate Disk Installation
[uxxs1354:root:/home/root:]
```

3. Implementation Plan for AIX Upgrade using NIMADM

In this phase we can done the migration from AIX 5.3 to 6.1 . Now we can preform the implementation plan in client server

3.1 Implementation Plan in Client server **Duration : 40 Min**

1. Run the pre_migration script **Duration : 25 Min**

Run this script, review the output (in /home/pre_migration), and correct any issues that it reports before migrating.

Our summary is available at:
/home/pre_migration.101008042454

Next we make available a disk with comparable size to rootvg. In this case we are using the disk used to mirror rootvg.

2 . Remove the efixes
Duration : 5 Min

Commit any applied fileset also remove any efixes that might hinder the migration process

instfix -r -L <fix>

lslpp -l | grep -i applied
installp -c

3. Enable rshd **Duration : 5 Min**

To allow NIMADM to do its job, we must temporarily enable rshd on the client LPAR .We will disable it again after the migration.

chsubserver -a -v shell -p tcp6 -r inetd
refresh -s inetd

```
root@ukcmuxxa329:/ # chsubserver -a -v shell -p tcp6 -r inetd
root@ukcmuxxa329:/ # refresh -s inetd
root@ukcmuxxa329:/ #
```

4. Re-create .rhosts file
Duration : 5 Min

Remove and re-create the .rhosts file under the root, give only owner permission.

#cd /
#rm .rhosts

```
root@ukcmuxxa329:/ # cd /
root@ukcmuxxa329:/ # rm .rhosts
root@ukcmuxxa329:/ #
```

#vi .rhosts
put " + " symbol and save it
(OR)
Put the " hostname " " root " and save it
chmod 600 .rhosts

```
root@ukcmuxxa329:/ # cat .rhosts
+
root@ukcmuxxa329:/ # chmod 600 .rhosts
```

Do the same process at the NIM server also and run the following command just check on client server **rsh** is enable or not .Both the server must be able to execute remote commands using **rsh**

rsh <nim server name> <command>

rsh uxxs1354   oslevel

5. Client server should have free disk which is equal or enough space for the NIMADM operation .This server having altinst_rootvg image on the hdisk1 ,remove it and use it for NIMADM operation.

3.2 Implementation plan in NIM server Duration : 200 Min

In this phase we can done the migration from AIX 5.3 to 6.1

1. Create a VG in NIM server Duration : 2 Min

Create one volume group with the name **migvg** to save the cache filesystems ,space of the volume group should be more than client's rootvg

bootinfo -s hdisk1
mkvg -y migvg -s 128 hdisk1

2. Enable rshd Duration : 5 Min

Enable the rshd temporarily. Remove the comment from the /etc/inetd.conf file. Disable after the migration

vi /etc/inetd.conf

```
##
ftp     stream  tcp6    nowait  root    /usr/sbin/ftpd          ftpd
telnet  stream  tcp6    nowait  root    /usr/sbin/telnetd       telnetd -a
shell   stream  tcp6    nowait  root    /usr/sbin/rshd          rshd
kshell  stream  tcp     nowait  root    /usr/sbin/krshd         krshd
login   stream  tcp6    nowait  root    /usr/sbin/rlogind       rlogind
#klogin  stream  tcp     nowait  root    /usr/sbin/krlogind      krlogind
#exec    stream  tcp6    nowait  root    /usr/sbin/rexecd        rexecd
#comsat dgram   udp     wait    root    /usr/sbin/comsat        comsat
#uucp   stream  tcp     nowait  root    /usr/sbin/uucpd         uucpd
bootps  dgram   udp     wait    root    /usr/sbin/bootpd        bootpd /etc/bootptab
##
```

refresh -s inetd

3. Once verify lpp_source and spot
Duration : 5 Min

Once verify the lpp_source and spot (Rstate = ready for use)
lsnim -l lpp_source_61-07-01

```
[uxxs1354:root:/home/root:] lsnim -l lppsource_61-07-01
lppsource_61-07-01:
   class       = resources
   type        = lpp_source
   arch        = power
   Rstate      = ready for use
   prev_state  = unavailable for use
   location    = /export/lpp_source/core/6.1/61-07-01
   simages     = yes
   alloc_count = 0
   server      = master
[uxxs1354:root:/home/root:]
```

lsnim -l spot_61-07-01

```
[uxxs1354:root:/home/root:] lsnim -l spot_61-07-01
spot_61-07-01:
   class          = resources
   type           = spot
   plat_defined   = chrp
   arch           = power
   Rstate         = ready for use
   prev_state     = verification is being performed
   location       = /export/spot/6.1/spot_61-07-01/usr
   version        = 6
   release        = 1
   mod            = 7
   oslevel r      = 6100-07
   alloc_count    = 0
   server         = master
   if_supported   = chrp.64 ent
   Rstate_result  = success
[uxxs1354:root:/home/root:]
```

4. bos.alt_disk_install is installed in rootvg for NIM server **Duration : 2 Min**

The same level of bos.alt_disk_install.rte must be installed in the rootvg (on NIM server) and in the SPOT which will be used.

lslpp -l bos.alt_disk_install.rte
#nim -o showres spot_name | grep -i bos.alt_disk_install.rte
#nim -o showres lpp_source | grep -i bos.alt_disk_install.rte

5. Add the bos.alt_disk_install.rte fileset **Duration : 5 Min**

If SPOT doesn't exits the bos.alt_disk_install.rte fileset add the fileset from lpp_source to SPOT

nim -o cust -a filesets = bos.alt_disk_install.rte -a lpp_source = <lpp_source name>   <spot_name>

nim -o showres spot_name | grep -i bos.alt_disk_install.rte

6. Run the NIMADM command **Duration : 3 Hrs**

On the NIM server run the nimadm command for migration

nohup nimadm -j migvg -c <client server name> -l lppsource_61-07-01 -s spot_61-07-01 -d hdisk1 -Y >> /tmp/nimadm_15May14 &

tail /tmp/nimadm_03Apr14

Where
- j specifies the VG name on the master
-c is the client server name
-l is the lpp_source name
-s is the spot name
-d is the Hdisk name for the alternate root VG (altinst_rootvg)
-Y agrees the license

Once migration starts cache filesystems create the in **migvg** and removes the after successful migration.

7.The migration process will takes in 12 phases .Migration is completed successfully, verify the altinst_rootvg is created on the client server.

#lspv

```
root@ukcmuxxa329:/ # lspv
hdisk0          00c9d10e3fd7e5a6                    rootvg          active
hdisk1          00c9d10e27feb91d                    altinst_rootvg  active
hdisk2          0002c7d1de871d00                    discovg         active
hdisk3          00c9d10e4a09bc61                    cln_findbavg    active
hdisk4          0002c7d13f87d993                    cln_fincdcvg    active
hdisk5          00c9d10e4a09bcd2                    cln_findbavg    active
hdisk6          00c9d10e4a09bd42                    cln_findbavg    active
hdisk7          00c9d10ed9ce827f                    fincdcvg        active
hdisk8          00c9d10ed9ce82fd                    fincdcvg        active
hdisk9          00c9d10ed9ce88ad                    fincdcvg        active
hdisk10         00c9d10ec210035b                    None
hdisk11         00c9d10ec21003fa                    None
hdisk12         00c9d10ec2100496                    None
```

4.Post Migration Duration : 120 Min

Client server OS Migration is successfully completed ,next we can do the post migration work on this server.

1. Clear look on NIMADM log Duration : 5 Min

Clear look at the nimadm log and check if there are any errors
/var/adm/ras/alt_mig

2. Stop the application and database
Duration : 20Min

Inform the Application and DATABASE team to bringing down the application and database

Stop application
Stop database

3. Change the bootlist
Duration : 2 Min

Set the primary boot as alt_install VG (aix 6.1) followed by old_rootvg

#bootlist -m normal -o
#bootlist -m normal hdisk1 hdisk0

4.Reboot the server
Duration: 20 Min

shutdown -Fr

5. Verify oslevel
Duration: 1Min

Verify the oslevel and check updated version

oslevel -s

```
root@ukcmuxxa329:/ # oslevel -s
6100-07-01-1141
root@ukcmuxxa329:/ #
```

6. Check if any missing filesets Duration : 15 Min

Check for any missing fileset on the client and try to install the same on the server to meet the expected level

lppchk -v m3
instfix -I | grep ML

7. Verify take performance and kernel parameters after migration Duration : 10 Min

Validate the outputs of performance and kernel parameters per-migration and post-migration. In AIX 6.1 many tunable parameters are hardcoded. If it required we have change the values manually and verify the below commands to validate the performance parametes.

lsattr -El sys0
vmo -a

7. Bringing up the Application and Database Duration : 15 Min

Bringing up DATABASE and Application server: Application & DB Team

Start Database
Start Application

Verify application is working fine or not

8. Disable the rshd Duration : 5 Min

9. Check TSM sheduler is running
Duration : 2 Min

ps -ef | grep dsmc

Application ,Database and all are working fine then remove the old_rootvg on hdisk0 . Next mirror the rootvg .

5. Backout Plan Duration : 90 Min

If the migration completed but issues are discovered after the reboot, then we can back out easily by booting from the original rootvg disk.

1. Alt disk roll back

2. Change the bootlist on the server to the old_rootvg disk (earlier version of AIX)
#bootlist -m normal -o

#bootlist -m normal hdiskX hdiskY

3. Reboot the server

shutdown -Fr

4. Check oslevel -s to ensure the server at it's earlier version

5. Run alt_disk_install -X to remove the alternate disk on the server

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